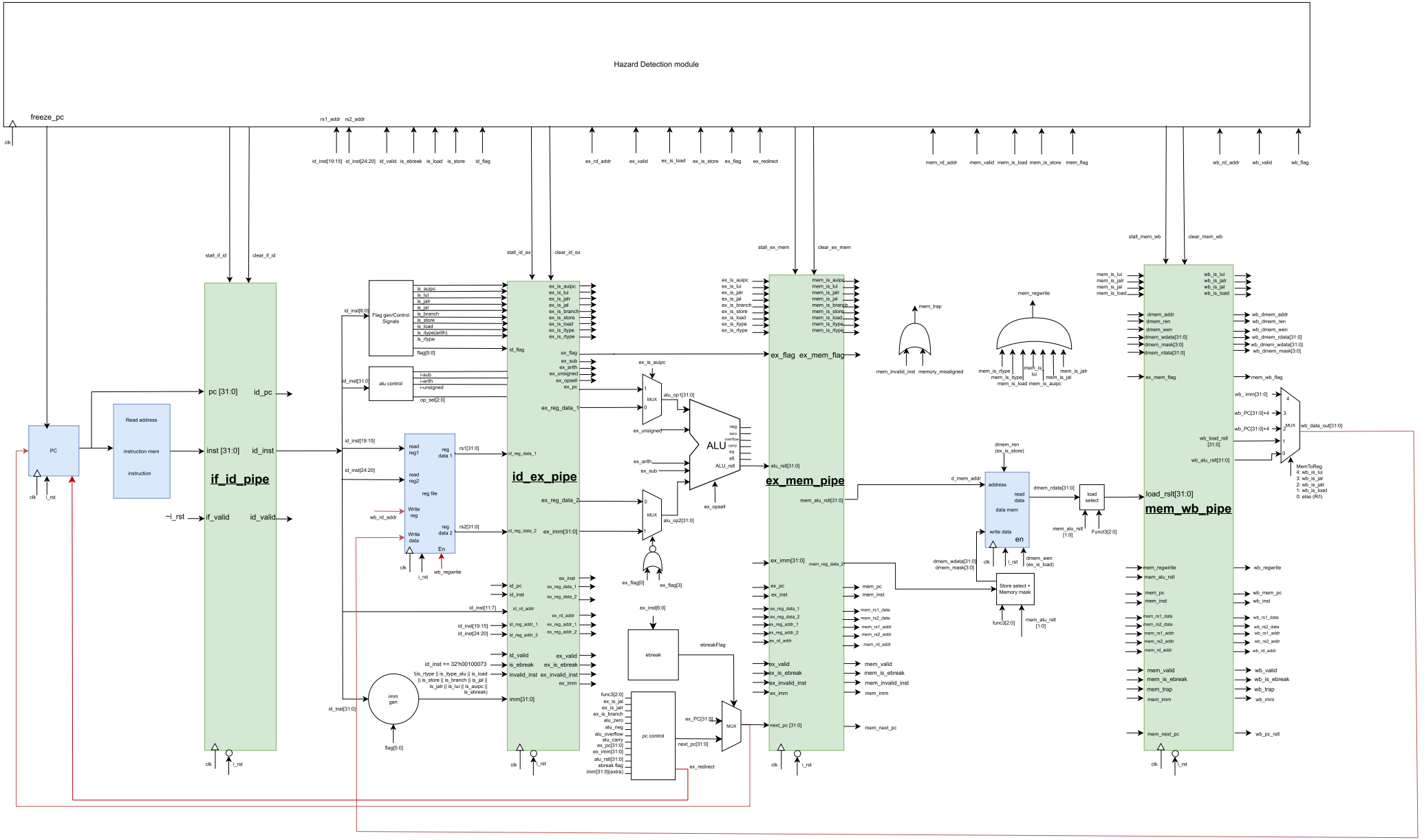

Hazard Detection module
freeze_pc
clk
rs1_addr rs2_addr
id_inst[19:15] id_inst[24:20] id_valid is_ebreak is_load is_store id_flag
ex_rd_addr ex_valid ex_is_load ex_is_store ex_flag ex_redirect
mem_rd_addr mem_valid mem_is_load mem_is_store mem_flag
wb_rd_addr wb_valid wb_flag
stall_if_id clear_if_id
stall_id_ex clear_id_ex
stall_ex_mem clear_ex_mem
stall_mem_wb clear_mem_wb
PC
Read address
instruction mem
instruction
pc [31:0] id_pc
inst [31:0] id_inst
if_id_pipe
~i_rst
if_valid id_valid
Flag gen/Control Signals
alu control
op_sel[2:0]
flag[5:0]
reg file
id_ex_pipe
ex_mem_pipe
mem_wb_pipe
ALU
ALU_rslt
MUX
imm gen
ebreak
ebreakFlag
pc control
next_pc[31:0]
ex_redirect
mem_trap
mem_invalid_inst
memory_misaligned
mem_regwrite
dmem_ren (ex_is_store)
d_mem_addr
address
read data
data mem
write data
en
dmem_wen (ex_is_load)
dmem_wdata[31:0]
dmem_mask[3:0]
dmem_rdata[31:0]
load select
Funct3[2:0]
Store select + Memory mask
load_rslt[31:0]
wb_data_out[31:0]
MemToReg
4: wb_is_lui
3: wb_is_jar
2: wb_is_jalr
1: wb_is_load
0: else (R/I)